

COURSE: DSA4050

COURSE TITLE: Deep Learning for Computer Vision

INSTRUCTOR: DR. EDWARD OMBUI

Week 11 Assignment: Unsupervised Learning in Computer Vision

Total Marks: 100

Submission Deadline: Monday 31st March 2025

Format: Jupyter Notebook/Python Script with Report

Objective

This assignment aims to assess your ability to apply **unsupervised learning techniques** in computer vision. You will work with an image dataset and implement different methods to extract meaningful representations, analyze patterns, and evaluate model performance.

Expected Learning Outcomes

By completing this assignment, you will:

- ✓ Understand the application of **unsupervised learning** in computer vision.
- ✓ Develop hands-on experience in **clustering, dimensionality reduction, and contrastive learning**.
- ✓ Analyze and compare different **representation learning** techniques.
- ✓ Enhance your ability to work with image datasets in **machine learning pipelines**.

Instructions

- Use **Python** and frameworks such as **NumPy, scikit-learn, PyTorch, and OpenCV**.
- Write clean, well-commented code.
- Submit a **Jupyter Notebook/Python script** with a short report explaining your methodology, results, and observations.
- Include visualizations wherever necessary.

Task 1: Image Data Preprocessing (10 marks)

1. Use the Road Crackimage dataset (check in the portal, week 11)
2. Perform necessary **data preprocessing**:

- Convert images to grayscale (if required).
- Normalize pixel values.
- Apply data augmentation techniques (e.g., flipping, rotation).

Deliverables:

- Code implementation
- Short explanation of preprocessing steps

Task 2: Clustering for Image Categorization (20 marks)

1. Extract features from images using Histogram of Oriented Gradients (HOG), ORB, or deep features from a pretrained CNN.
2. Apply K-Means and DBSCAN clustering to group similar images.
3. Use the Silhouette Score to evaluate clustering quality.

Deliverables:

- Code for feature extraction and clustering
- Visualization of clusters (e.g., sample images from each cluster)
- Evaluation and interpretation of results

Task 3: Dimensionality Reduction and Visualization (20 marks)

1. Apply Principal Component Analysis (PCA) and t-SNE to reduce image feature dimensions.
2. Visualize the results in 2D plots.

Deliverables:

- Code for PCA and t-SNE
- Scatter plots with labeled clusters
- Discussion on differences in PCA vs. t-SNE results

Task 4: Contrastive Learning for Representation Learning (30 marks)

1. Implement SimCLR or another contrastive learning approach to learn visual representations.
2. Train the model using contrastive loss and visualize the learned representations using t-SNE.
3. Evaluate the learned representations by training a simple classifier on top of them.

Deliverables:

- Code for SimCLR implementation
 - t-SNE visualization of learned representations
 - Comparison of classifier accuracy with vs. without contrastive learning
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Task 5: Evaluation and Comparative Analysis (20 marks)

1. Compare the performance of **clustering-based vs. contrastive learning-based** representations.
2. Analyze which method is more effective for image categorization and why.
3. Provide a **3-4 paragraph report** summarizing your findings, including strengths and limitations of each approach (do this on the Jupyter notebook). The report should include **methodology, key observations, and performance comparisons**.

Deliverables:

- Comparative analysis (tables/graphs)
- Report discussing insights and observations

Bonus Task (Optional, +10 marks)

1. Explore a **GAN-based unsupervised learning approach** (e.g., **DeepCluster, SelfGAN**) for feature learning.
2. Compare its performance with SimCLR.

Deliverables:

- Code implementation
- Brief analysis of results

Grading Criteria

Task	Marks
Image Data Preprocessing	10
Clustering (K-Means, DBSCAN)	20
Dimensionality Reduction (PCA, t-SNE)	20
Contrastive Learning (SimCLR)	30
Evaluation & Analysis	20

	Task	Marks
Total		100